

Mohamed Morchid

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Associate Professor in Computer Science

Research interests

Speech and Language Processing, Spoken Language Understanding, Machine Learning, Semantic Web.

Education

Ph.D. in computer science **2011 – 2014**
University of Avignon - LIA *(38 months and 4 days)*

M.Sc. in computer science **2009 – 2011**
University of Avignon - CERI *(24 months)*

Professional experiences

Associate Professor, University of Avignon **Since 2015**
Research: Speech and Language Processing.

Teaching and research assistant (ATER), LIA-Language **2014 – 2015**
Research: Spoken Language Understanding, contNomina project.

Ph.D. researcher and lecturer, LIA-Language **2011 – 2014**
Research: Robust representation of noisy documents (spoken, web,...).

Awards, fellowships and grants

Qualification **2015**
from the French CNU 27 (informatics)

Ph.D. degree with honors **2014**
from University of Avignon - LIA

ISCA, International Speech Communication Association Grant **2014**
for the INTERSPEECH 2014 conference

Ph.D. fellowship **2011 – 2014**
of the French National Research Agency (ANR-10-CORD-007)

M.Sc. degree with honors **2011**
from the University of Avignon - CERI

Professional societies

IEEE, The Institute of Electrical and Electronics Engineers

ISCA, International Speech Communication Association

AFCP, Association Francophone de la Communication Parlée

Ph.D. thesis

Title: *Robust Representation of noisy documents in homogeneous spaces*

Defended: November 25th 2014 front of the jury consisting of:

Jury President:

Professor Frédéric BÉCHET, University of Aix Marseille, Marseille, France

Reviewers:

Apple Distinguished Scientist Jérôme R. BELLEGARDA, Apple Inc., Cupertino, USA

Professor Laurent BESACIER, Université of Joseph Fourier, Grenoble, France

Professor François YVON, Université Paris-Sud, Paris, France

Examiners:

Senior Scientist Youssef HAMADI, Microsoft Research, Cambridge, UK

CNRS Scientist Benjamin PIWOWARSKI, LIP6, Paris, France

Advisers:

Professor Georges LINARÈS, University of Avignon, Avignon, France

Assistant Professor Richard DUFOUR, University of Avignon, Avignon, France

Abstract: In the Information Retrieval field, a document is usually considered as a "bag-of-words". This model does not take into account the temporal structure of the document and is sensitive to noises which can alter its lexical form. These noises can be produced by different sources: uncontrolled form of documents in micro-blogging platforms, automatic transcription of speech documents which are error-prone, lexical and grammatical variabilities in Web forums... The work presented in this thesis addresses issues related to document representations from noisy sources. My thesis consists of three parts in which different representations of content are available. The first one compares a classical representation based on a term-frequency representation to a higher level representation based on a topic space. The abstraction of the document content allows us to limit the alteration of the noisy document by representing its content with a set of high-level features. The major problem with this topic-based representation is that its parameters are chosen empirically. The second part presents a novel representation based on multiple topic spaces that allow us to solve three main problems: the closeness of the subjects discussed in the document, the tricky choice of the "right" values of the topic space parameters and the robustness of the topic-based representation. Based on the idea that a single representation of the contents cannot capture all the relevant information, we propose to increase the number of views on a single document. This multiplication of views generates "artificial" observations that contain fragments of useful information. However, it has the disadvantage of being very large, redundant and of containing additional variability associated with the diversity of views. I have proposed a method based on factor analysis to compact different views to obtain a new robust representation of low dimension which contains only the informative part of the document while the noisy variabilities are compensated. Nonetheless, during the learning process of topic spaces, the document is considered as a "bag-of-words" while many studies have showed that the word position in a document is useful. A representation which takes into account the temporal structure of the document based on hyper-complex numbers is proposed in the third part. This representation is based on the hyper-complex numbers of dimension four named quaternions.

Keywords: Robust representation, noisy document, latent Dirichlet allocation, factor analysis.

M.Sc. thesis

Title: *Buzz prediction*

Defended: *June 25th 2011*

Reviewer:

Professor Yannick ESTÈVE, Université of Le Mans, France

Adviser:

Professor Georges LINARÈS, Université of Avignon, France

Abstract: The aim of the internship was to propose a bursty events prediction method. This one has been applied in the context of the Twitter platform. The prediction is performed by a neural network that combines three indices in order to predict the number of retweets. These indices are related to popularity, expressivity and singularity. The popularity index is based on the analysis of RSS streams. The expressivity uses a dictionary that contains words annotated in terms of expressivity load. The singularity represents outlying topic association estimated with a Latent Dirichlet Allocation (LDA) model. I have also participated to the MediaEval'11 SED benchmark.

Keywords: *Bursty event, latent Dirichlet allocation, Social Event Detection, MediaEval.*

Ph.D. thesis mentored

★ **Kilian Janod, (U of Avignon)** : Provide original features from **possibility theory** for speech processing and propose original methods to embed possibility in language models for speech recognition, co-supervised with Dufour R. and Lianrès G., 2013 - ...

★ **Mohamed Bouaziz, (U of Avignon)** : The purpose of this thesis titled "Delinearization of synchronous video streams" is to find out original methods to combine different streams to predict a sequence of TV shows, co-supervised with Dufour R. and Lianrès G., 2013 - ...

Interns mentored

★ **Amina Brahem, (ENSI of Tunis)** : The aim of the project is to identify the letter displayed on a screen seen by the user, by using her or his **EEG** signals only. Multi-user context is taken into account in order to build a letter recognition system suitable for any human being, co-supervised with Lianrès G., 2014, 3 months, Now: Ph.D. student at U of Tunis

★ **Etienne Papegnies, (U of Avignon)** : The aim is to manage a basic interface using brain **EEG** signals only, co-supervised with Lianrès G., 2014, 3 months, Now: Ph.D. student at U of Avignon

★ **Mathias Quillot, (U of Avignon)** : Use of context for proper name recognition in diachronic audio documents. In the context of diachronic data (data which change over time), co-supervised with Dufour R. and Lianrès G., 2014, 2 months, Now: Master student at U of Avignon

International research collaborations

★ **ACL CICLing 2015** : The number of documents available into Internet moves each day up. For this reason, processing this amount of information effectively and expressibly becomes a major concern for companies and scientists. Methods that represent a textual document by a topic representation are widely used in Information Retrieval (IR) to process big data such as **Wikipedia** articles. One of the main difficulty in using topic model on huge data collection is related to the material resources (CPU time and memory) required for model estimate. To deal with this issue, we proposed to build topic spaces from summarized documents. We have

presented a study of topic space representation in the context of big data. The topic space representation behavior is analyzed on different languages. Experiments show that topic spaces estimated from summaries are as relevant as those estimated from the complete documents. The real advantage of such an approach is the processing time gain: we showed that the processing time can be drastically reduced using summarized documents (more than 60% in general). This study finally points out the differences between thematic representations of documents depending on the targeted languages such as English or latin languages.

Participants: Mohamed Morchid (France), Juan-Manuel Torres-Moreno (Canada, Mexico, France), Richard Dufour (France), Javier Ramirez-Rodriguez (Mexico) and Georges Linarès (France).

★ **ACM CrowdMM 2014** : This work concerns human computation approaches for social image labeling. In particular, the work studies how crowdsourcing-based approaches to image tag validation can help with the use of human input from the crowd, in the form of votes collected from workers on a crowdsourcing platform. Indeed, validating user tags helps to refine them, making them more useful for finding images. In the case of interpretation-sensitive tags, however, automatic (i.e., pixel-based) approaches cannot be expected to deliver optimal results. Instead, human input is the key. Experiments in the domain of social fashion images are carried out using the dataset published by the Crowdsourcing Task of the [Mediaeval 2013 Multimedia Benchmark](#). Experimental results reveal that when a larger number of crowd-contributed votes are available, it is difficult to beat a majority vote. However, additional information sources, i.e., crowdworker history and visual image features, allow us to maintain similar validation performance while making use of less crowd-contributed input. In short, validation of interpretation-sensitive user tags for social images is possible, with “just a little help from the crowd.”

Participants: Babak Loni (Netherlands) Delft University, Jonathon Hare (United Kingdom) University of Southampton, Mihai Georgescu (Germany) L3S Hannover, Michael Riegler (Norway) Simula Research Laboratory, Xiaofei Zhu (Germany) L3S Hannover, **Mohamed Morchid** (France), University of Avignon, Richard Dufour (France), University of Avignon and Martha Larson (Netherlands) Delft University.

Visiting scientist

Title: *The PowerPoint Assistant*

Period: from 11/2013 to 01/2014 (3 months)

Adviser: [Professor Youssef Hamadi](#)

Abstract: I spent time visiting [Microsoft Research, Cambridge](#) (UK) Laboratory to work under the supervision of [Professor Youssef Hamadi](#). The PowerPoint Assistant project's aim is to provide relevant content to Microsoft customers. Thus, I used original methods to allow Microsoft customers to enrich their PowerPoint presentations with relevant content. This project was selected to be showed at annual Microsoft meeting TechFest'14 (40 projects were submitted and 22 were selected). During this meeting, Microsoft Research scientists from across the globe gather to display and discuss their latest projects, which encompass a broad spectrum of computer-science investigations. A demo is available [here](#).

Organization committee

[EACL](#), European Chapter of the Association for Computational Linguistics (ACL)

2012

Projects

GaFes (under contract ANR-14-CE24-0022) <i>Participants: LIA, Centre Norbert Elias, Syllabs, GECE, EURECOM</i>	2014 – 2018
ContNomina (under contract ANR-12-BS02-0009) <i>Participants: LORIA, LIA</i>	2013 – 2015
DECODA (under contract ANR-09-CORD-005) <i>Participants: LIF, RATP, SONEAR, LIA</i>	2012 – 2015
SuMACC (under contract ANR-10-CORD-007) <i>Participants: LIA, Wikio, Syllabs, EURECOM</i>	2011 – 2014

Lectures and courses (computer science)

University of Avignon - CERI, France <i>Various courses in Computer Science (368 hours detailed here):</i> ○ Java Programming ○ Object modeling ○ Object Oriented Programming ○ Internet engineering ○ ...	2011 – 2015
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Computer skills

Programming: Java, C++, C#, Python

Research: MatLab, R

Languages

Native French Speaker and Intermediate English Speaker.

Selected publications

- **Morchid, M.**, Dufour, R. and Linarès, G. "Impact of Word Error Rate on Theme Identification Task of Highly Imperfect Human-human Conversations." In *Computer Speech and Language*, Elsevier Edition, 2016.
- **Morchid, M.**, Bouallegue, M., Dufour, R., Linarès, G., Matrouf, D. and De Mori, R. "Compact Multiview Representation of Documents based on the Total Variability Space." In *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2015.
- **Morchid, M.**, Dufour, R., Bousquet, P.-M, Linarès, G. and J.-M. Torres-Moreno, "Feature Selection using Principal Component Analysis for Massive Retweet Detection." In *Pattern Recognition Letters*, Elsevier ed., 2014.

Publications

International Journals

1. **Morchid, M.**, Dufour, R. and Linarès, G. "Impact of Word Error Rate on Theme Identification Task of Highly Imperfect Human-human Conversations." In *Computer Speech and Language, Elsevier Edition*, 2016.
2. **Morchid, M.**, Bouallegue, M., Dufour, R., Linarès, G., Matrouf, D., and De Mori, R. "Compact Multiview Representation of Documents based on the Total Variability Space." In *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2015.
3. **Morchid, M.**, Dufour, R., Bousquet, P.-M., Linarès, G., and Torres-Moreno, J.-M. "Feature Selection using Principal Component Analysis for Massive Retweet Detection." In *Pattern Recognition Letters, Elsevier Edition*, 2014.

International Conferences

1. **Morchid, M.**, Dufour, R., and Linarès, G. "Topic-Space based Setup of a Neural Network for Theme Identification of Highly Imperfect Transcriptions." In *International Automatic Speech Recognition and Understanding Workshop (ASRU)*, 2015.
2. **Morchid, M.**, Josselin, D., Dufour, R., Altman, E., and Linarès, G. A Topic Modeling based Representation to Detect Tweet Locations. " In *the International Society for Photogrammetry and Remote Sensing (ISPRS Geospatial week)*, 2015.
3. **Morchid, M.**, Dufour, R., and Matrouf, D. "A comparison of normalization techniques applied to latent space representations for speech analytics." In *Conference of the International Speech Communication Association (INTERSPEECH)*, 2015.
4. **Morchid, M.**, Portilla, Y., Josselin, D., Dufour, R., Altman, E., El-Beze, M., Cossu, J.-V., Linarès, G., and Reiffers-Masson, A. "An author topic-based approach to cluster tweets and mine their location." In *International Conference on Spatial Statistics: Emerging Patterns (SPATIAL)*, 2015.
5. **Morchid, M.**, Dufour, R., Linarès, G., and Hamadi, Y. "Latent topic model based representations for a robust theme identification of highly imperfect automatic transcriptions." In *International Conference on Intelligent Text Processing and Computational Linguistics (CICLing)*, 2015.
6. **Morchid, M.**, Torres-Moreno, J.-M., Dufour, R., Rodriguez, J., and Linarès, G. "Automatic summarization approaches to speed up topic model learning process." In *International Conference on Intelligent Text Processing and Computational Linguistics (CICLing)*, 2015.
7. Estève, Y., Bouallegue, M., Lailier, C., **Morchid, M.**, Dufour, R., Linares, G., Matrouf, D., and De Mori, R. "Integration of word and semantic features for theme identification in telephone conversations." In *International Workshop Series on Spoken Dialogue Systems Technology (IWSDS)*, 2015.

8. **Morchid, M.**, Dufour, R., Bouallegue, M., and Linarès, G. "Author-topic based representation of call-center conversations." In *International Spoken Language Technology Workshop (SLT)* , 2014.
9. Loni, B., Hare, J., Georgescu, M., Riegler, M., Zhu, X., **Morchid, M.**, Dufour, R., and Larson, M. "Getting by with a little help from the crowd: Practical approaches to social image labeling." In *International Workshop on Crowdsourcing for Multimedia (CrowdMM)*, 2014.
10. **Morchid, M.**, Bouallegue, M., Dufour, R., Linarès, G., Matrouf, D., and De Mori, R. "An i-vector based approach to compact multi-granularity topic spaces representation of textual documents." In *the Conference of Empirical Methods on Natural Language Processing (EMNLP)*, 2014.
11. **Morchid, M.**, Dufour, R., and Linarès, G. "A combined thematic and acoustic approach for a music recommendation service for tv commercials." In *International Society for Music Information Retrieval Conference (ISMIR)*, 2014.
12. **Morchid, M.**, Dufour, R., Bouallegue, M., Linarès, G., and De Mori, R. "Theme identification in human-human conversations with features from specific speaker type hidden spaces." In *Conference of the International Speech Communication Association (INTERSPEECH)*, 2014.
13. **Morchid, M.**, Bouallegue, M., Dufour, R., Linarès, G., Matrouf, D., and De Mori, R. "I-vector based representation of highly imperfect automatic transcriptions." In *Conference of the International Speech Communication Association (INTERSPEECH)*, 2014.
14. Bouallegue, M., **Morchid, M.**, Dufour, R., Driss, M., Linarès, G., and De Mori, R. "Subspace gaussian mixture models for dialogues classification." In *Conference of the International Speech Communication Association (INTERSPEECH)*, 2014.
15. Bouallegue, M., **Morchid, M.**, Dufour, R., Driss, M., Linarès, G., and De Mori, R. "Factor analysis based semantic variability compensation for automatic conversation representation." In *Conference of the International Speech Communication Association (INTERSPEECH)*, 2014.
16. **Morchid, M.**, Dufour, R., Niaz, U., Bouvier, F., de Groc, C., Linarès, G., Meiraldo, B., and Peralta, B. "Sumacc project's corpus: A topic-based query extension approach to retrieve multimedia documents." In *Proceedings of the 17th International Conference on Text, Speech and Dialogue (TSD)*, 2014.
17. **Morchid, M.**, Dufour, R., and Linarès, G. "A lda-based topic classification approach from highly imperfect automatic transcriptions." In *Proceedings of the 9th edition of the Language Resources and Evaluation Conference (LREC)*, 2014.
18. **Morchid, M.**, Linarès, G., and Dufour, R. "Characterizing and predicting bursty events: The buzz case study on twitter." In *Proceedings of the 9th edition of the Language Resources and Evaluation Conference (LREC)*, 2014.
19. **Morchid, M.**, Dufour, R., Bousquet, P.-M., Bouallegue, M., Linarès, G., and De Mori, R. "Improving dialogue classification using a topic space representation and a gaussian classifier based

- on the decision rule." In *International Conference on Acoustic, Speech and Signal Processing (ICASSP)*, 2014.
20. **Morchid, M.**, Dufour, R., and Linarès, G. "Thematic representation of short text messages with latent topics: Application in the twitter context." In *Conference of the Pacific Association for Computational Linguistics (PACLING)*, 2013.
 21. **Morchid, M.**, Linarès, G., El-Beze, M., and De Mori, R. "Theme identification in telephone service conversations using quaternions of speech features." In *Conference of the International Speech Communication Association (INTERSPEECH)*, 2013.
 22. **Morchid, M.** and Linarès, G. "Lda-based method for automatic tagging of youtube videos." In *International Workshop on Image and Audio Analysis for Multimedia Interactive Services (WIAMIS)*, 2013.
 23. **Morchid, M.**, Dufour, R., and Linarès, G. "Event detection from image hosting services by slightly-supervised multi-span context models." In *International Workshop on Content-Based Multimedia Indexing (CBMI)*, 2013.

National Conferences

1. **Morchid, M.**, Dufour, R., and Linarès, G. "Initialisation de Réseaux de Neurones à l'aide d'un Espace Thématique." In *Conférence sur le Traitement Automatique des Langues Naturelles (TALN)*, 2015.
2. Janod, K., **Morchid, M.**, Dufour, R., and Linarès, G. "Apport de l'information temporelle des contextes pour la représentation vectorielle continue des mots." In *Conférence sur le Traitement Automatique des Langues Naturelles (TALN)*, 2015.
3. Lailler, C., Estève, Y., De Mori, R., Bouallegue, M., and **Morchid, M.** "Utilisation d'annotations sémantiques pour la validation automatique d'hypothèses dans des conversations téléphoniques." In *Conférence sur le Traitement Automatique des Langues Naturelles (TALN)*, 2015.
4. **Morchid, M.**, Dufour, R., Linarès, G., and De Mori, R. "Classification de transcriptions automatiques imparfaites : Doit-on adapter le calcul du taux d'erreur-mot ?" In *Proceedings Journées d'étude de la parole (JEP)*, 2014.
5. **Morchid, M.** and Linarès, G. "Prédiction des buzz sur twitter." In *CONFérence en Recherche d'Information et Applications (CORIA)*, 2013.
6. **Morchid, M.**, Dufour, R., and Linarès, G. "Combinaison de thèmes latents pour la contextualisation de tweets." *Atelier Contextualisation de Messages Courts (EGC)*, 2013.
7. **Morchid, M.** and Linarès, G. "Extraction de mots-clefs dans des vidéos web par analyse latente de Dirichlet." In *Proc. Journées d'étude de la parole (JEP)*, 2012.

International Workshops

1. **Morchid, M.**, Huet, S., and Richard, D. "A topic-based approach for post-processing correction of automatic translations." In *International Workshop on Spoken Language Translation (IWSLT)*, 2014.
2. **Morchid, M.**, Dufour, R., Bouallegue, M., Linarès, G., and Matrouf, D. "Lia@Mediaeval 2013 Crowdsourcing task: Metadata or not metadata? that is a fashion question." In *(MediaEval)*, 2013.
3. **Morchid, M.**, Dufour, R., Bouallegue, M., Linarès, G., and Matrouf, D. "Lia @ mediaeval 2013 musiclef task: A combined thematic and acoustic approach." In *(MediaEval)*, 2013.
4. Bouallegue, M., Senay, G., **Morchid, M.**, Matrouf, D., Linarès, G., and Dufour, R. "Lia @ mediaeval 2013 spoken web search task: An i-vector based approach." In *(MediaEval)*, 2013.
5. Cossu, J.-V., Bigot, B., Bonnefoy, L., **Morchid, M.**, Bost, X., Senay, G., Dufour, R., Bouvier, V., Torres-Moreno, J.-M., and El-Bèze, M. "Lia@replab 2013." In *An evaluation campaign for Online Reputation Management Systems (CLEF)*, 2013.
6. Bost, X., Brunetti, I., Cabrera-Diego, L. A., Cossu, J.-V., Linhares, A., **Morchid, M.**, Torres-Moreno, J.-M., El-Bèze, M., and Richard, D. "Systèmes du lia à deft'13." In *Actes du neuvième Défi Fouille de Textes (DEFT)*, 2013.
7. **Morchid, M.** and Linarès, G. "Inex 2012 benchmark a semantic space for tweets contextualization." In *Workshop of the Initiative for the Evaluation of XML Retrieval (CLEF)*, 2012.
8. **Morchid, M.** and Linarès, G. "Mediaeval benchmark: Social event detection using lda and external resources." In *(MediaEval)*, 2011.

Invited talks & presentations

1. Bouaziz, M., **Morchid, M.**, Richard, D. and Lianrès G. "Structuration des Flux TV : Etat de l'Art." In *Journée commune AFIA - ARIA 2015 : Intelligence artificielle et Recherche d'Information*, 2015.
2. **Morchid, M.** and Linarès, G. "Bursty event prediction by content-based approach." In *Third Workshop of the Brazilian Institute for Web Science Research*, 2012.